

Farm Management System

A Comprehensive Salesforce-Based Farm Management & Crop Tracking System

1. Introduction

Digitizing Agricultural Records

Project Overview

- **FARMSYNC** is an application built on Salesforce for recording, organizing, and reviewing crop lifecycles and farm resource usage.
- It uses custom objects, connected records, dashboards, and automation features to present agricultural data in a highly structured manner.
- The application allows agronomists, farm managers, and administrators to maintain statistics, prepare reports, and use data to support farming decisions.
- Converts traditional, fragmented paper-based operational logs into a secure, centralized cloud format.



Purpose & Objectives

Core Purpose

This project aims to completely overhaul how farm data is collected, validated, and utilized, transitioning from error-prone manual methods to a streamlined digital ecosystem.

By empowering agricultural professionals with real-time data, farms can optimize resource usage, improve yield predictions, and ultimately drive better profitability.

Key Objectives

- Convert crop and resource performance records into a secure digital format.
- Make agricultural statistics more accurate, transparent, and easier to access remotely.
- Provide timely performance analysis to assist with seasonal planning.
- Reduce manual data handling and errors through powerful Salesforce automation.

2. Ideation Phase

Understanding the User Needs

Problem Statements

ID	Target Customer	Primary Goal	Current Obstacle	Root Cause
PS-1	Farm Manager	Track crop yields across different plots.	Cannot get accurate statistics easily.	Data is stored in different physical files and formats.
PS-2	Team Foreman	Maintain complete worker & resource records.	Takes too much time to update and verify data.	Most daily logs are maintained manually on paper.
PS-3	Agronomist	Analyze seasonal trends & generate reports.	Faces difficulty comparing historical performances.	There is no centralized data analytics system.

Target Users



Farm Managers

Need dependable daily operations data and clearer insights to make critical scheduling and resource allocation decisions.



Agronomists

Require accurate historical crop statistics and advanced analytics tools to research soil health and optimize future yields.



System Admins

Responsible for securely configuring the CRM platform, managing user access roles, and maintaining overall data integrity.

Empathy Map: Think & Feel

Think & Feel

- Need highly accurate crop data to drive profitability.
- Need precise resource usage statistics (water, fertilizer).
- Require much better analytics tools than standard spreadsheets.
- Feel anxious when data is delayed or inconsistent across reports.

Hear & See

- **Hear:** Market demands require decisions to be highly data-driven to remain competitive.
- **Hear:** Directives from ownership to streamline operations.
- **See:** Multiple, conflicting data sources and disconnected legacy systems.
- **See:** Scattered spreadsheets causing confusion.

Empathy Map: Pains & Gains

! Pain Points

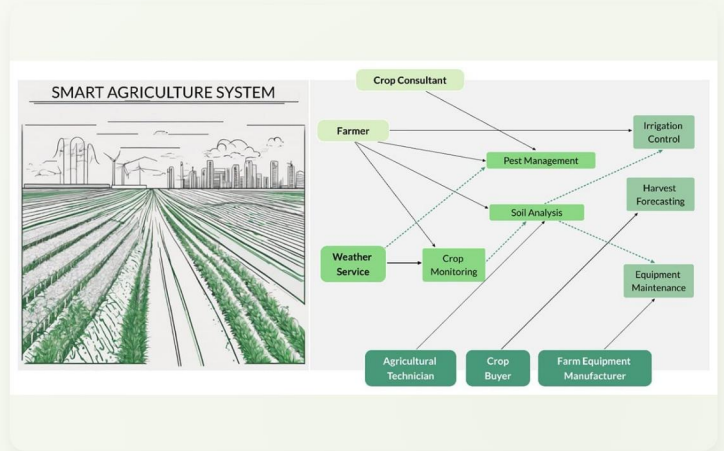
- Dealing with completely disconnected legacy systems.
- Frequent data entry errors and frustrating reporting delays.
- Inconsistent crop statistics across different departmental reports.
- Hours wasted manually verifying information before it can be used.
- Hard to trust the accuracy of the final data presented to stakeholders.

✓ Desired Gains

- One centralized, reliable analytics tool for the entire farm operation.
- Automated, accurate reports generated at the click of a button.
- Quick, data-driven insights to guide immediate field actions.
- Faster decision-making capabilities regarding harvesting schedules.
- Reduction in tedious, manual administrative tasks.

Brainstorming Ideas

- During the ideation phase, several concepts were evaluated to solve the farm's data fragmentation issues.
- **Idea 1:** A basic web-based crop management system (too limited).
- **Idea 2:** A standalone mobile statistics tracker (lacked robust backend analytics).
- **Idea 3:** AI-based performance prediction software (too complex for initial adoption).
- **Selected Idea:** A comprehensive, Salesforce-integrated CRM platform (FARMSYNC) combining cloud storage, mobile accessibility, and powerful dashboard analytics.



3. Requirements Analysis

Defining System Capabilities

Customer Journey



Functional Requirements

FR No.	Core Requirement Description
FR-1	The system must allow users to register and maintain detailed crop profiles.
FR-2	Users must be able to enter and safely save daily harvest and resource details.
FR-3	The system must display performance information through interactive dashboards.
FR-4	Authorized users must be able to create custom reports from stored crop data.
FR-5	The application must automate routine calculations using Salesforce Flow.

NFR: Usability & Security

NFR-1: Usability

FARMSYNC must be straightforward for all users to operate, regardless of technical background.

- Provide a clean, user-friendly Lightning interface.
- Keep system navigation clear and highly consistent.
- Ensure dashboards are legible and visually intuitive.
- *Example:* A new field worker can easily log daily harvest totals without needing specialized training.

NFR-2: Security

Farm data must be strictly protected so only authorized users can view or modify records.

- Assign access privileges based on the user's specific role.
- Require secure, multi-factor user authentication.
- Protect stored information with enterprise-grade encryption.
- *Example:* Field workers can log data, but only Admins can delete or alter historical configurations.

NFR: Reliability & Performance

NFR-3 & 5: Reliability & Availability

The application must operate consistently, preserving data accuracy and remaining accessible.

- Prevent accidental loss of vital operational records.
- Perform automated, regular cloud backups.
- Aim for 24/7 access throughout the year, especially during peak harvest seasons.
- *Example:* If a user loses internet in the field, offline caching prevents data loss until reconnected.

NFR-4 & 6: Performance & Scalability

The system must respond promptly and adapt as the farm's data volume grows exponentially.

- Load Lightning pages and dynamic dashboards without delay.
- Keep response times low when querying large datasets.
- Accommodate significant growth in user numbers and data points.
- *Example:* Reports analyzing 10 years of crop data across 50 plots must render within seconds.

Data Flow Overview

Information moves systematically through the FARMSYNC application to ensure integrity and usability:

- **Data Ingestion:** Live field data (harvest yields, resource usage) and crop profiles are inputted via the Salesforce UI or Mobile App.
- **Data Storage:** Records are securely stored within custom Salesforce Objects (Crop, Harvest).
- **Processing:** Salesforce Flows automate calculations, such as total seasonal yields and resource averages.
- **Output:** The processed data is instantly rendered into dynamic Dashboards and Reports for management review.



Technology Stack

System Layer	Chosen Technology	Primary Function
Front End (UI)	Salesforce Lightning	Delivers a responsive, accessible interface for web and mobile users.
Analytics	Salesforce Reports & Dashboards	Visualizes complex crop data into easy-to-read charts and metrics.
Automation	Salesforce Flow	Handles complex business logic, validations, and automated data updates.
Back End	Salesforce Custom Objects	Defines the relational database structure for crops, plots, and harvests.
Database/Storage	Salesforce Cloud	Ensures secure, scalable, and highly available data hosting.

4. Project Design Phase

Architecting the Solution

Problem–Solution Fit



Centralized Database

Problem: Scattered, inconsistent paper records.

Solution: Keeps all crop and match data in a single, highly organized repository.



Automated Collection

Problem: Manual tracking is tedious.

Solution: Replaces manual math with automated roll-ups and real-time interface updates.



Data–Driven Insights

Problem: Uncertainty in planning.

Solution: Eliminates guesswork by providing advanced comparison tools and accurate trend reports.

Solution Architecture

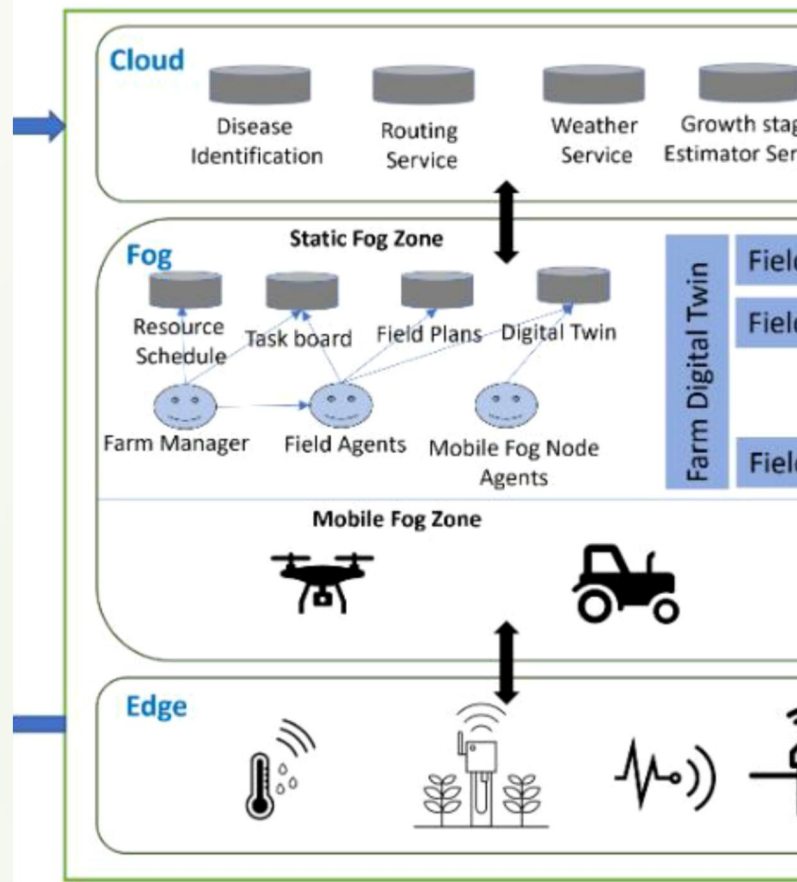
Four-Tiered Cloud Design

Presentation Layer: Web dashboards and responsive mobile applications built on Lightning UI facilitate seamless data entry from the field.

Application Layer: Embedded business logic and APIs process analytics and enforce strict data validation rules.

Data Layer: Secure cloud storage houses all relational data, protecting historical records and maintaining organizational hierarchy.

Analytics Layer: Dedicated reporting engines process raw storage into trend visualizations and custom performance reports.



5. Planning & Scheduling

Agile Execution

AGILE EXECUTION TABLES

User Stories & Backlog

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	Member 1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	Member 2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	Member 3
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.	3	High	Member 1
Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that complete performance details can be captured accurately.	5	Medium	Member 1
Sprint-2	Data Structure	USN-6		5	High	Member 1

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			As a user, I want customized page layouts so that only relevant fields are displayed during data entry.			
Sprint-2	Automation	USN-7	As a system, I want to validate player and match inputs so that incorrect data entries are prevented.	3	High	Member 3
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	Member 4

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	2 Days	21 SEP 2026	23 SEP 2026	25	27 SEP 2026
Sprint-2	20	2 Days	21 SEP 2026	23 SEP 2026		
Sprint-3	20	2 Days	21 SEP 2026	24 SEP 2026		
Sprint-4	20	2 Days	21 SEP 2026	24 SEP 2026		

6. Development Phase: System Implementation & Project Flow

System Implementation Overview

The FARMSYNC system implementation relies on a robust architecture of custom objects and automated flows to manage agricultural data.

Crop Batch Object (Parent): This object tracks the overarching details and lifecycle of a specific planted crop. Key fields include Crop Name, Season Type, Expected Yield, and Status. It utilizes roll-up summaries to track Total Actual Yield, Total Water Used, and Total Harvests Logged, acting as the primary anchor for field activity data.

Harvest Log Object (Child): This records individual daily or weekly harvest logs and is tied directly to a specific crop batch via a Master-Detail relationship. Key fields include Harvest Date, Yield Quantity, Quality Grade, and Field Worker.

Automation & Flows: The system uses Roll-Up Calculations to automatically aggregate daily harvest yields and resource inputs to the parent Crop Batch level. Validation Rules enforce data quality at the point of entry, preventing future dates for completed harvests or impossibly high yield numbers. Screen Flows guide field workers through a step-by-step data entry process on their mobile devices to reduce errors.

Project Flow Milestones

- **Milestone 1:** Salesforce Developer Account Creation
- **Milestone 2:** Object Creation
- **Milestone 3:** Tabs
- **Milestone 4:** The Lightning App
- **Milestone 5:** Fields & Relationships
- **Milestone 6:** Field Dependency
- **Milestone 7:** Profiles and Roles
- **Milestone 8:** User Creation
- **Milestone 9:** Sharing Rules
- **Milestone 10:** Record Creation
- **Milestone 11:** Reports
- **Milestone 12:** Dashboards

Step-by-Step Implementation

Milestone 1 - Salesforce developer account creation

Activity 1: Creating Developer Account

1. Go to <https://developer.salesforce.com/signup>
2. On the sign-up form, enter the First name & Last name, Email, Role (Developer), Company (College Name), Country (India), Postal Code (pin code), and a unique Username (e.g., username@organization.com).
3. Click on sign me up.

Activity 2: Account Activation

1. Go to your email inbox, find the verification email (may take 5-10 mins), and click Verify Account.
2. Provide a password, answer a security question, and click change password to redirect to the Salesforce setup page.

Milestone 2 - Object Creation

Activity 1: Create Farm Object

1. From setup click on object manager. Click create, select custom object.
2. Set Label to "Farm", Plural label to "Farms", and Record name to "Farm Name".
3. Select data type as "Text".
4. Enable Allow Reports, Track Field History, and Allow Search. Ensure Deployment Status is Deployed.
5. In Object Creation Options, select Add Notes and Attachments related list to default page layout. Click Save.

Activity 2: Create Crop Object

1. Click the gear icon, select Setup, and go to the Object Manager tab. Click Create and select Custom Object.
2. Set Label to "Crop", Plural Label to "Crops", and Record Name to "Crop Name".
3. Select data type as "Text".
4. Check Allow Reports, Allow Search, and Add Notes and Attachments related list to default page layout. Click Save.

Activity 3: Create Buyer Object

1. From the Object Manager, click Create and select Custom Object.
2. Set Label to "Buyer", Plural Label to "Buyers", and Record Name to "Buyer Name".
3. Select data type as "Text".
4. Check Allow Reports, Allow Search, and Add Notes and Attachments related list to default page layout. Click Save.

Milestone 3 - Tabs

Activity 1: Custom Tab Creation

1. **Farm Tab:** From Home, enter Tabs in Quick Find. Under custom object tabs, click New. Select Farm, choose any icon, and click Next, Next, Save.
2. **Crop Tab:** Repeat the process, selecting Crop as the object.
3. **Farmer Tab:** Repeat the process, selecting Farmer as the object.
4. **Buyer Tab:** Repeat the process, selecting Buyer as the object.

Milestone 4 - The Lightning App

Activity 1: Create a Lightning App

1. From Setup, enter App Manager in Quick Find and select App Manager. Click New Lightning App.
2. Enter "Farm Management System" as the App Name, click Next. Leave App Options and Utility Items as default.
3. From Available Items, select Farm, Crop, Farmer, Buyer, Reports, and Dashboards and move them to Selected Items.
4. From Available Profiles, select System Administrator, move it to Selected Profiles, and click Save & Finish.

Milestone 5 - Fields & Relationships

Activity 1: Create Custom Fields for Farm Object

1. In Object Manager, select Farm, go to Fields & Relationships, and click New.
2. Create Picklist "Farm Type" (Values: Crop, Livestock, Mixed).
3. Create Picklist "Soil Quality" (Values: Organic, Cover Crops, Composting).
4. Create Picklist "Irrigation type" (Values: Sprinkler irrigation, Drip irrigation, localized irrigation).
5. Create Text field "Location".
6. Create Text field "Size".
7. Create a Master-Detail relationship with the Farmer Object.

Activity 2: Create Custom Fields for Crop Object

1. Create Text field "Crop Name".
2. Create Date field "Planting Date".
3. Create Date field "Harvest date".
4. Create a Lookup relationship to the Farm Object.
5. Create Currency field "Actual Yield".
6. Create Number field "Total Acres Harvested" (Maximum Length 20).
7. Create Formula field "Expected Yield per Acre" (Return type: Currency) with the formula:
Actual Yield / Total Acres Harvested.

Activity 3: Create Custom Fields for Farmer Object

1. Create Phone field "Contact Number".
2. Create Picklist "State" (Values: Telangana, Andhra Pradesh, Tamilnadu).
3. Create Picklist "District" (Enter corresponding district values).
4. Create Picklist "Village" (Enter corresponding village values).

Activity 4: Create Custom Fields for Buyer Object

1. Create Phone field "Contact".
2. Create Email field "Email".
3. Create Text area "Address".

4. Create Picklist "Buyer Type" (Values: Wholesale, Retail).

Milestone 6 - Field Dependency

Activity 1: Creation of field dependency (Farmer Object)

1. In the Farmer object under Fields & Relationships, click Field Dependency and then New.
2. Select Controlling Field as State and Dependent Field as District.
3. Map the appropriate values by double-clicking them (e.g., Telangana - Yadadri Bhongir).
4. Click Include Values, repeat for other states, test via Preview, and Save.

Milestone 7 - Profiles and Roles

Activity 1: Create Profiles

1. From setup, enter profiles in Quick Find. Select Standard user and click clone.
2. Name it "Sales Department" and save.
3. Clone the Standard user profile again, name it "Farm Manager", and save.

Activity 2: Create Roles

1. Enter roles in Quick Find and select set up roles.
2. Below CEO, click add role, enter label "Farm Manager", and save.

Milestone 8 - User Creation

Activity 1: Create Users

1. From Setup, enter Users in Quick Find. Click New User.
2. Enter First name "Farm", Last name "Manager", along with email and a unique username.
3. Select role "Farm Manager", user License "Standard Platform User", and profile "Farm Manager". Save.
4. Create another user named "Sales Department" with the Sales Manager profile.

Milestone 9 - Sharing Settings

Activity 1: Create Sharing Rules

1. Go to Sharing Settings in Quick Find.
2. Under the Crop object, click New sharing rule. Give it a label.
3. Select the Crop field as equal to True.
4. Share with roles: Farm Manager, setting access to Read/Write. Save.
5. Create a similar Sharing Rule to share Buyer records with the Sales department (Read/Write access).

Milestone 10 - Record Creation

Activity 1: Create Records

1. Click the App Launcher and open the Farm Management system.
2. Navigate to the Farmer Tab, click New, fill in the details, and Save.

Milestone 11 - Reports

Activity 1: Create Reports

1. Open the Reports tab from the App Launcher. Click New Report and select "Farmer with Farms". Click start report.
2. In Outline, select columns to display. In Group, select fields to group by.
3. Save the report as "Farmer with Farms" in a public folder.
4. Repeat to create reports for Crops and Buyers.

Milestone 12 - Dashboards

Activity 1: Create Dashboards

1. Click the Dashboard tab and select New Dashboard. Name it "Farmers with farms" and create.
2. Click +Component, select the "Farmers with Farms" report, and choose the Gauge chart. Click Add, Save, and Done.
3. Verify the records by navigating to the Dashboard Tab within the Farm Management System app.

7. Conclusion & Future

Looking Ahead

Future Scope Enhancements



Offline Mobile App

Develop a robust, offline-capable mobile application allowing field workers to log resources from remote farm locations with seamless syncing upon reconnection.



AI Yield Prediction

Incorporate artificial intelligence to analyze historical crop data alongside live weather patterns, predicting future performance and optimizing harvest timing.



IoT Sensor Integration

Connect soil moisture hardware and weather sensors directly into Salesforce, providing real-time irrigation monitoring and automated smart agricultural alerts.